

## Fungal polysaccharopeptide inhibits tumor angiogenesis and tumor growth in mice

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### Abstract

Angiogenesis is crucial to tumor growth and metastasis, and interruption of this process is a prime avenue for therapeutic intervention of tumor proliferation. The present study has made use of the S180 tumor-bearing mouse model to investigate the polysaccharopeptide, PSP, isolated from the edible mushroom *Coriolus versicolor*, a herbal medicine known for its anti-angiogenesis properties. Quantitative analysis of microcorrosion casting of the tumor tissue showed more angiogenic features such as dense sinusoids and hot spots, in control (untreated) than in PSP-treated animals. Immunostaining of tumor tissues with antibody against the endothelial cell marker (Factor VIII) demonstrated a positive correlation in that both the vascular density and tumor weight were lower in mice treated with PSP. Morphometric analysis of corrosion casts revealed that, even though the total amount of new vessel production was reduced, the basic tumor type-specific vascular architecture was retained. However, the expression of vascular endothelial cell growth factor (VEGF) in these tumors was suppressed. In conclusion, anti-angiogenesis should be one of the pathways through which PSP mediated its anti-tumor activity.

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### Introduction

Angiogenesis is the formation of new blood vessels from pre-existing vessels. This process is most active in embryogenesis, and in certain pathological conditions such as tumor progression (Folkman,

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1995). As the tumor continues to grow, it requires more oxygen and nutrients from surrounding blood vessels which infiltrate into the tumor. Since angiogenesis is crucial for nourishment and progression of most solid tumors (Folkman, 1972) and their metastases, anti-angiogenesis has long been considered a prime avenue for cancer therapy (Dvorak et al., 1995; Fan et al., 1995). Tumor angiogenesis involves the interaction of tumor cells, endothelial cells, and extracellular matrix, as well as changes in the balance of angiogenic and anti-angiogenic factors. One of the most potent angiogenic factors is vascular endothelial cell growth factor, VEGF, which is expressed in many types of developing solid tumors (Boockock et al., 1995; Ferrer et al., 1997; Hashimoto et al., 1995; Yamaguchi et al., 1998). Antibodies against VEGF inhibit tumor growth through the inhibition of tumor angiogenesis (Fennie et al., 1995; Ferrara, 1999). Up-regulation of VEGF is often found in tumor cells, both in cell lines and in solid tumors (Boockock et al., 1995). A number of studies have indicated that the expression of VEGF and its receptors is up-regulated by hypoxia (Iizuka et al., 1994) which often occurs in the inner regions of the tumor, particularly in tumor cells next to the necrotic region (Plate et al., 1994). VEGF not only stimulates the proliferation of endothelial cells (Shibuya, 1995), but also increases vascular permeability (Roberts and Palade, 1997; Shibuya, 1995), and induces the release of von Willebrand factor (Factor VIII) from Weibel-Palade bodies by endothelial cells (Brock et al., 1991). VEGF also acts as a chemotactic factor for peripheral blood monocytes (Clauss et al., 1990).

In recent years, various experimental models have been proposed for the study of angiogenesis. In vivo methods include the rabbit ear chamber, hamster cheek pouch, rat corneal micropocket, and chick embryo chorioallantoic membrane. In vitro methods utilize cultures of human or animal endothelial cells from macro- or micro-vessels (for review, see Auerbach et al., 1991). Although endothelial cell culture provides a useful tool for studying cell migration, proliferation and secretion (Querrec et al., 1993), with the heterogeneity of endothelial cell subpopulations from different sources, the method has its drawback in providing confusing information at times (Haraldsen et al., 1996; Hashimoto et al., 1995; Swerlick et al., 1992). Moreover, in vitro cultures may not provide all of the factors and cells that are present in the in vivo microenvironment (Pintavorn and Ballermann, 1997).

The present study was undertaken to investigate the fungal herb, *Coriolus versicolor*, specifically its polysaccharopeptides which are known to exhibit anti-tumor activities. Our aim was to determine whether the inhibition was mediated through its anti-angiogenesis properties, and to elucidate the possible mechanism(s) of action in an in vivo model. The dual focus of the study was to examine the qualitative and quantitative changes of the microvasculature of PSP-treated animals, by comparing them with untreated controls, and to assess the status of VEGF expression which is known to play an important role in angiogenesis.

## Materials and methods

### *Preparation of polysaccharopeptide (PSP) solution*

The polysaccharopeptide (PSP), isolated from deep-layer cultivated mycelia of *Coriolus versicolor* (Liu et al., 1993), has a molecular weight of 100 KDa. The protein content of the PSP solution was measured by the method of Bradford (1976), and a stock solution of 20 mg protein/ml was made with distilled water and was diluted to appropriate concentrations when used.

## *Animals*

One hundred inbred male mice (BALB/c mice, 6–8 weeks old) were used in this study. All the mice were supplied by the Laboratory Animal Service Center of The Chinese University of Hong Kong. The animals were housed under normal laboratory conditions ( $21 \pm 2$  °C, 12/12 h light/dark cycle) and maintained on standard rodent chow.

Murine sarcoma S180 cells ( $1 \times 10^6$  cells) (ATCC TIB 66, USA) were implanted subcutaneously in the back of each mouse. Five days after inoculation, 50 mice were fed with the PSP solution in their drinking water (35 µg/day/mouse) (Liu et al., 1993). The other 50 mice received only drinking water and served as controls. For simplicity, the PSP-treated group was referred as “treated group”. Starting from Day 9 until Day 29, 12 mice from each group were killed every five days for immunohistochemical, morphological and gene expression analyses. The weights of the tumors were also recorded for comparison.

The procedures involving animals and their care were conducted in conformity with the institutional guidelines that are in compliance with national and international laws and policies (UK Animals Act: Scientific Procedures, 1986; NIH guide for the care and use of laboratory animals, NIH publication no. 85-23, 1985).

## *Microvascular corrosion casting*

Corrosion casts of the tumor vascularity were made in both control and treated animals on day 15 as described earlier (Konerding et al., 1995). Briefly, the animals were anesthetized with chloral hydrate (7%, 0.5 ml/100 g body weight i.p.), and the casting medium, Mercor CL-2B(r) (02686-AA, SPI Supplies, USA) in methylmetacrylate (v/v: 4/1), was used injected into the vascular system after perfusion fixed with warm glutaraldehyde solution (2.5% phosphate-buffered glutaraldehyde, pH 7.4). The tumors were removed after hardening in warm water ( $\sim 40$  °C), and the tumor tissues were corroded in 10% KOH solution at 40 °C. The casts were washed with distilled water, frozen, dried in a lyophilizer, and mounted on specimen stubs using thin copper wires as conductive bridges (Lametschwandtner and Lametschwandtner, 1992). The casts were coated with palladium-gold for stereo pairs observation using a JEOL scanning electron microscope (JSM 6301F, JEOL, Japan) with a magnification between  $25\times$  and  $500\times$  at an acceleration voltage of 3 kV and a working distance of at least 15 mm.

## *Morphometry of microvascular architecture*

The 3-dimensional microvascular architecture was reconstructed from the images of four randomly selected corrosion casts from both control group and treated group with an image analysis program (Kontron KS 300, Echting, Germany), and the inter-vessel and inter-branch distances of the vascular network were analyzed (Malkusch et al., 1995). In total, 1905 inter-vessel and 955 inter-branch distances were calculated. In addition, the vascular diameters and the variation of the diameter in individual vessel segments were assessed in 509 micro-vessels.

Statistical analyses and graphic displays were performed using SigmaStat and SigmaPlot (Jandel Scientific Software, Erkrath, Germany). Controls and treated tumors were tested for their frequency distribution before analysis. The differences in the frequency distributions were tested for signifi-

cance using the Chi-square test, whereas testing for significant differences using the mean and standard deviations of the distribution, was performed with the Mann-Whitney rank sum test and ANOVA.

### *Immunohistochemical staining*

#### *Preparation of paraffin sections*

Tumors excised from the mice were fixed in 4% paraformaldehyde in 0.1 M phosphate-buffer, pH 7.4, at room temperature overnight, dehydrated, embedded in paraffin wax and sectioned at 7  $\mu$ m thick before staining with a polyclonal antibody against von Willebrand Factor (Factor VIII, Dako, Denmark) for 3 hr at room temperature, followed by secondary antibody IgG conjugated to horseradish peroxidase (Tago, USA), and visualized using AEC chromogen (Zymed, USA). The sections were counterstained with haematoxylin and mounted in glycerol. A negative control was prepared for each tissue from the same block, with normal horse serum substituted for the primary antibody.

#### *Image analysis*

Image analysis was performed according to the method described by Kohlberger et al. (1996) using an imaging system consisting of a cooled color digital (CCD) camera (Spot, Universal Imaging Corp., USA) and a light microscope (Axiophot, Zeiss, Germany). Under the microscope, the four most highly vascularized areas in a section, known as 'hot spots' (Weidner et al., 1993), each of 0.25 mm<sup>2</sup>, were chosen at 100  $\times$  magnification for evaluation. Any red-staining endothelial cell or endothelial-cell cluster that was clearly separated from adjacent microvessels, tumor cells, and other connective-tissue elements was considered as a single, countable microvessel. Vessel lumina and red blood cells, although frequently present, were not necessary for consideration as markers for microvessels (Weidner et al., 1993).

Evaluation of the captured image was done using Metamorph software (Chinetek Scientific). Positively stained blood vessels were outlined manually. The number of vessels was counted and the area enclosed by the vessels was calculated. Eighteen and fifteen sections from the control group and the treated group, respectively, were examined. The figures were then used to calculate vascular density in number of blood vessels per 1 mm<sup>2</sup> and also in % area enclosed by the blood vessels. The relationship between tumor growth and vascular density was assessed.

#### *Gene expression study*

The tumors were homogenized in RNA extraction buffer on ice and total cellular RNA was isolated according to the method of Chomczynski and Sacchi (1987). All the reagents for the reverse-transcription polymerase chain reaction (RT-PCR) were purchased from Life Technologies, USA. Oligo(dT)-primed cDNAs were prepared in a volume of 20  $\mu$ l from 1  $\mu$ g of total RNA using SUPERScript™ II RNase free reverse transcriptase. PCR was carried out in a 25  $\mu$ l mixture using the PCR Reagent System. The primer used to amplify the cDNA fragments of VEGF was: forward 5' -GTGATCAAGTTCATGGACGTC-TACCAGCG-3' ; reverse 5' -TACGTTTCGTTTAACTCAAGCTGCCTCGC-3' (Fennie et al., 1995). For internal control,  $\beta$ -actin was employed by using the primers: forward 5' -CCCTCTACAATGAGC-3' ; reverse 5' -ACGTCACACTTCATG-3' (Scheuermann and Bauer, 1993). PCR was carried out in a 25

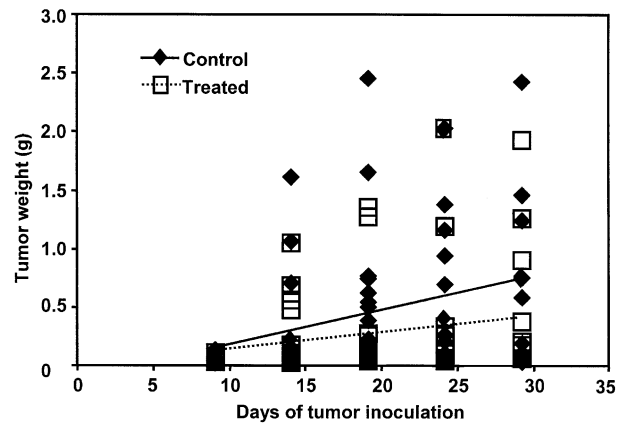


Fig. 1. Linear regression of tumor weights of both control and treated groups versus number of days after inoculation with S180 sarcoma cells.

$\mu$ l mixture using the PCR Reagent System (Life Technologies, USA) and 30 and 40 cycles of exponential amplification were performed for  $\beta$ -actin and VEGF respectively. Both genes were amplified with initial denaturation at 94 °C for 5 min and final extension at 72 °C for 7 min.

The amplified products were separated and analyzed by 1.2% (w/v) agarose gel electrophoresis, and their intensities were quantified by the software ImageQuant and normalized with  $\beta$ -actin.

### Statistical analysis

The Mann-Whitney test was used to assess the significance in the difference in body weight and tumor weight between the control and the treated group. In this test, the difference between the data of two different groups is considered significant when the exact significance (E.S.) is greater than the asymptotic significance (A.S.), both of which are generated by the software.

The Pearson's test was used to evaluate the significant correlation between (i) the number of vessels per unit area and the tumor weight, and (ii) the area of vessels per unit area with the tumor weight. Correlation is considered significant when  $p < 0.001$ . All computations were done with SPSS Base 8.0 statistical software (USA).

Table 1

Tumor weights of both control and treated groups after inoculation with S180 sarcoma cells from 9–29 days ( $n = 10$ )

| Day | Control group<br>(Mean $\pm$ SEM) | Treated group<br>(Mean $\pm$ SEM) |
|-----|-----------------------------------|-----------------------------------|
| 9   | 0.0629 $\pm$ 0.0129               | 0.0477 $\pm$ 0.0116               |
| 14  | 0.1901 $\pm$ 0.0607               | 0.2005 $\pm$ 0.0441               |
| 19  | 0.4065 $\pm$ 0.1066               | 0.2197 $\pm$ 0.0781               |
| 24  | 0.4178 $\pm$ 0.1341               | 0.2372 $\pm$ 0.0683               |
| 29  | 0.7393 $\pm$ 0.2036               | 0.3862 $\pm$ 0.1769               |

## Results

### *Tumor growth*

Fig. 1 and Table 1 show the tumor weight of both control and treated groups at various time points. The tumor growth in the control group was significantly greater than that in the treated group at each time point (E.S. > A.S.). The mean tumor weight on Day 29 was about 48% larger in the control than in the treated group.

### *Vessel morphology and distribution*

#### *Microcorrosion casting*

The tumor was induced by subcutaneous inoculation of mouse S180 sarcoma cells into the mice. In the subcutaneous space, the vasculature between the tumor border and normal host tissue represented the transition from normal non-proliferating host vessels to tumor vessels generated in

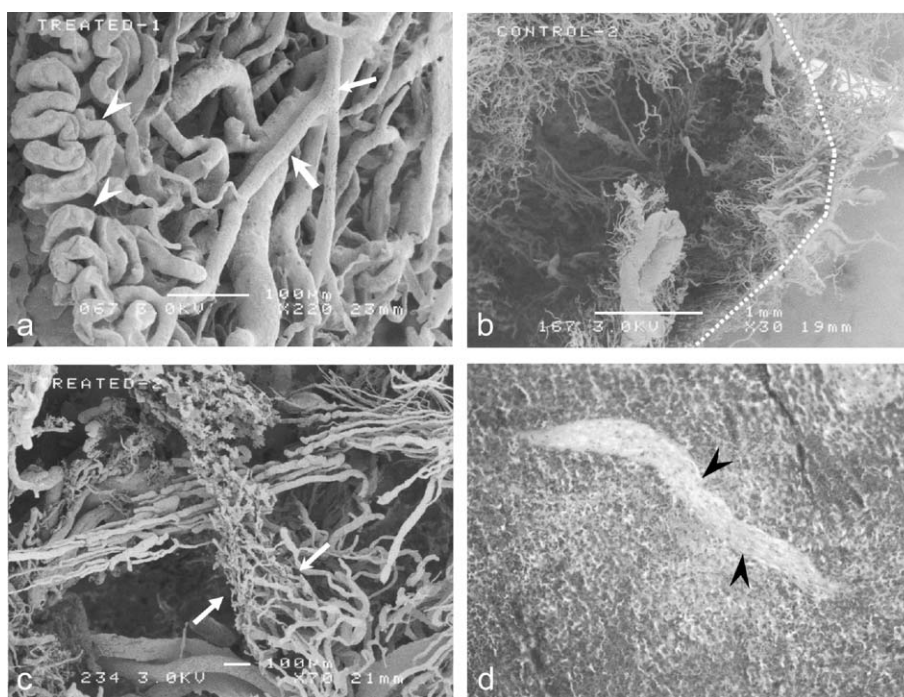


Fig. 2. (a) Periphery of a tumor. Here is shown the transition of normal blood vessels to tumor vessels. The normal vessels are characterized by an even diameter, a straight course, and a normal branching pattern (arrows). Further to the left of the figure are the typical tumor vessels, with dilation and meandering courses (arrowheads). (b) Periphery of a tumor. The blood vessels at the periphery form a vascular envelope (outlined by dotted line). (c) Vasculature of dying muscles (arrows) within tumor tissues. The blood vessels are less regular and regression is seen. This may result from nutrient deprivation of the muscles by the rapidly growing tumor. (d) The muscle fibre (arrowheads) wrapped by the tumor cells has shrunk in size and is dying, as revealed by paleness of the eosin stain (200 $\times$ ).



response to secreted, diffusible angiogenic growth factors (e.g. VEGF). Normal vessels had an even diameter and a normal hierarchy, but became dilated with meandering courses as they extended to the tumor (Fig. 2a) and formed a vascular envelope (Grunt et al., 1986) (Fig. 2b). The expanding tumors also enclosed some of the muscle fibers, which were eventually deprived of nutrition by the rapidly growing tumor and degenerated, as shown by the regressed vessels of the muscle (Fig. 2c–d).

The tumor vasculature displayed a highly variable network of vessels of varying densities. Densely vascularized regions alternated with avascular zones (Fig. 3a). This observation was confirmed in tumor sections immunostained with antibody against Factor VIII (Fig. 3b). At the core of the tumor, the vasculature appeared to be highly disorganized, characterized by an abundance of sinusoids (Fig. 3c) and glomeruloid vessels (Fig. 3d).

In comparison with tumors of the control group, those treated with PSP displayed a vasculature of few blood vessels, and much less density. A few dilated blood vessels were observed at the periphery of the tumor and the tumor vascular envelope was much less developed than that in the control group. At the

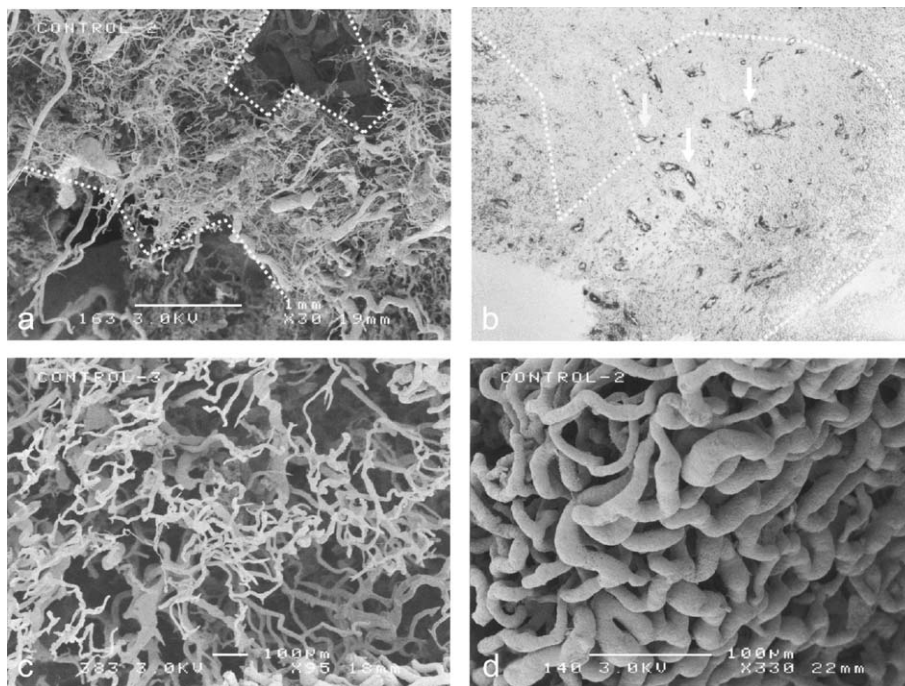


Fig. 3. (a) At the core of the tumor. Tumor vasculature is characterized by its highly varying density. The central portion of the vasculature is of rather high density and surrounded by comparatively avascular areas. (b) Highly vascular density showing positive Factor VIII immunostaining endothelia (arrows) is also demonstrated by light microscopy. In this figure, a densely vascular region (enclosed by dotted line) is surrounded by a comparatively avascular region (100 ×). (c) Tumor sinusoids. At the core of the tumor, a dense network of tumor sinusoids resulting from active angiogenesis can be seen. They serve to nourish the rapidly growing tumor. At higher magnification, they are usually flattened, resulting from the compression of the adjacent tumor cells. (d) Glomeruloid structure. These densely packed, coiled vessels are known as ‘hot spots’ of the tumor in immunostained sections. These regions also correspond to active angiogenesis.

core of the tumor, in the treated group, sinusoids or glomeruloid structures were also observed but in a much sparsely distributed pattern (Fig. 4a–b).

Morphometric analysis of the 3D casts revealed that the basic vascular architecture was largely retained despite the qualitative differences between the two groups of animals. (Fig. 5A–C) illustrate the vascular parameters which define the network as cumulative frequencies, allowing for assessment the distribution, median value, the minimum and maximum values, and possible differences between groups.

Fig. 5A shows the distribution of the intervessel distances plotted on a logarithmic scale, with the data most closely approximate to a log normal distribution. The median of the treated group is 41.4  $\mu\text{m}$ , about 10% smaller than that of the controls (45.0  $\mu\text{m}$ ;  $p = 0.0026$ ). However, the distribution curve shows, at least in the lower part, little difference in the intervessel distances. No statistically significant difference in interbranching distance was revealed between control and treated groups (Fig. 5B). Although the vessels of the controls (median = 17.4  $\mu\text{m}$ ) had in general a wider diameter than those of the treated tumors (median = 15.5  $\mu\text{m}$ ;  $p = 0.0015$ ), the heterogeneity of vessel diameters within individual vessel segments (not shown) did not show statistically significant difference.

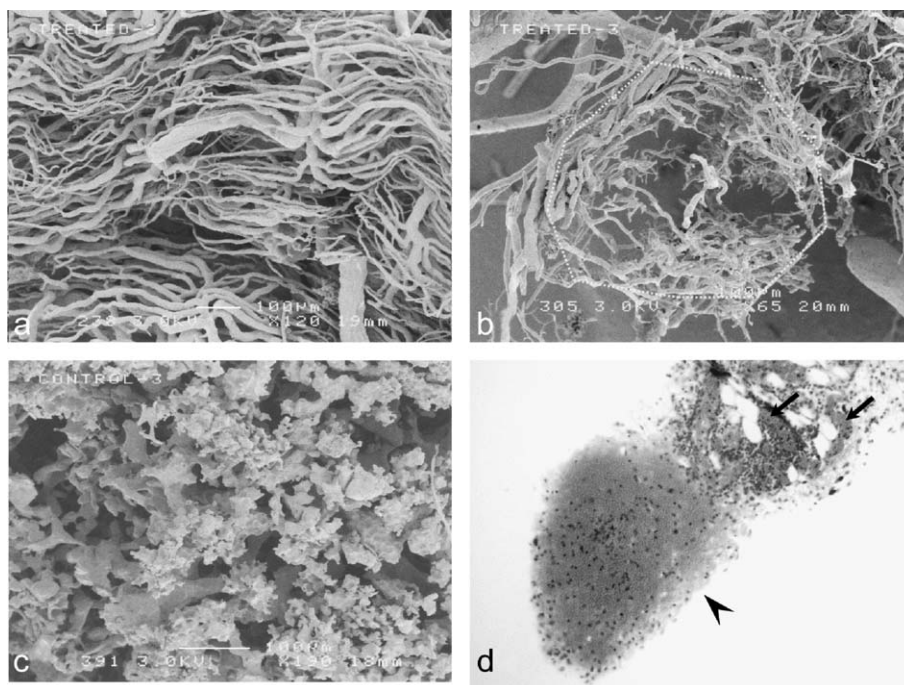


Fig. 4. (a) Subcutaneous blood vessels at the periphery of a tumor from the treated group. At high magnification, it can be seen that the vasculature resembles that of normal blood vessels. (b) Nodule-like structure (enclosed by dotted line) corresponding to active angiogenic area. (c) Extravasates from leaky vessels. (d) Light microscopic image of intact (arrows) and leaky blood vessels (arrow head) where red blood cells dispersed among the tumor cells (200 $\times$ ).



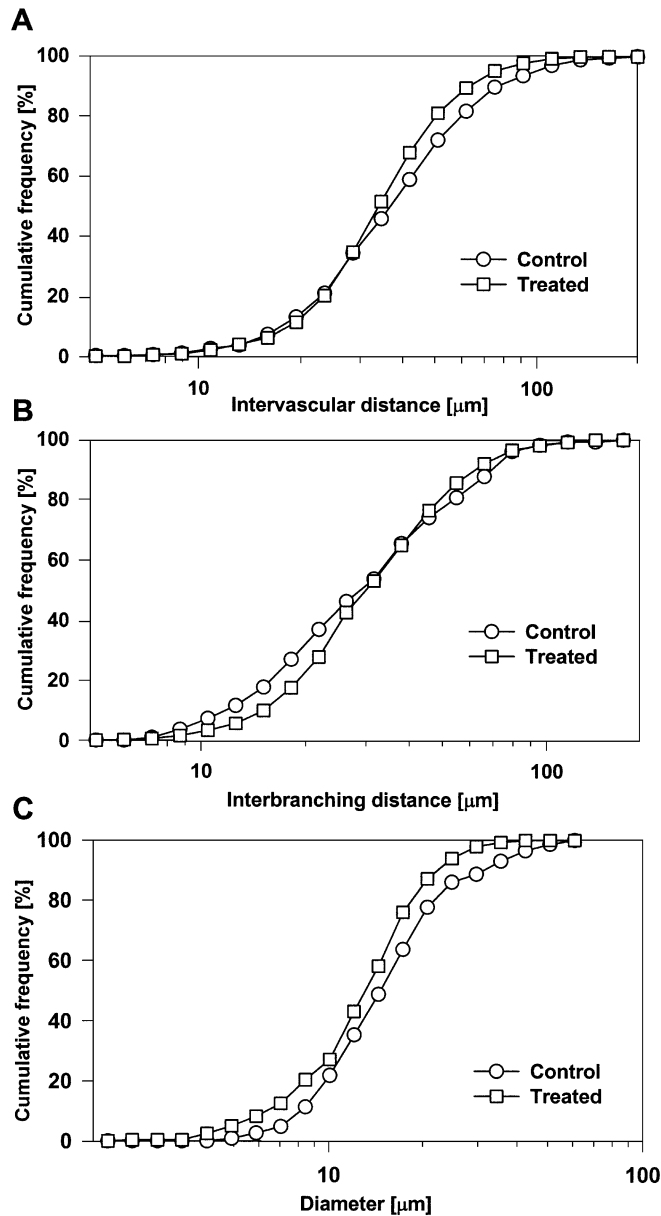


Fig. 5. A comparison of (A) interspace distances, (B) interbranching distances, and (C) vessel diameter between control and treated groups.

#### Immunohistochemical staining

A positive linear relationship was observed when the number of vessels counted was plotted against the tumor weight (Fig. 6). By the Pearson's test, the correlation of the number of vessels per unit area versus the tumor weight was considered significant ( $p < 0.001$ ). Both the blood vessel count and the tumor weight were higher in the control group (Fig. 6).

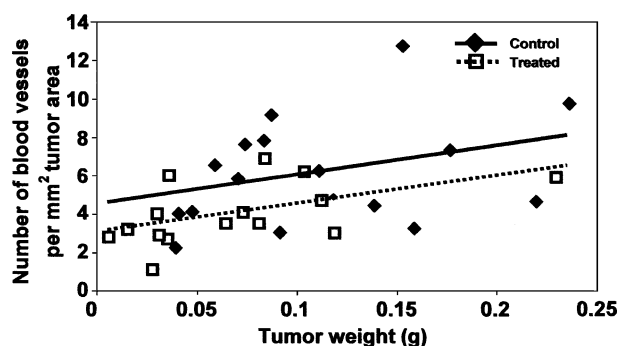


Fig. 6. The number of blood vessels per mm<sup>2</sup> tumor area versus tumor weight.

The density of the blood vessels was not significantly ( $p > 0.001$ ) correlated with the tumor weight, because their distribution was more diverse (data not shown).

### Gene expression study

The level of VEGF expression in the control group was generally much higher than that of the treated group (Fig. 7). In both groups, VEGF gene expression increased along with tumor progression, but the level of expression was consistently lower in the treated group.

## Discussion

Angiogenesis is an important process underlying tumor progression, and thus tumor growth inhibition mediated by anti-angiogenic agents is likely to be accompanied by significant changes in the tumor vasculature and the expression of angiogenic genes. The use of microcorrosion casting study, a technique used to great advantage in tracking the vascular change of a great variety of tumors such as melanoma, colon adenocarcinoma, Lewis lung carcinoma, angiosarcoma and leiomyosarcoma (Konerding et al., 1995), has brought forth an array of features which were different between the untreated and the PSP-treated groups. In the untreated group, cast images showed very active neovascularization, consisting of a dense network of tumor sinusoids in the core of the tumor, meandering vessels at the periphery, and leaky vessels. It is known that VEGF causes fenestrations in

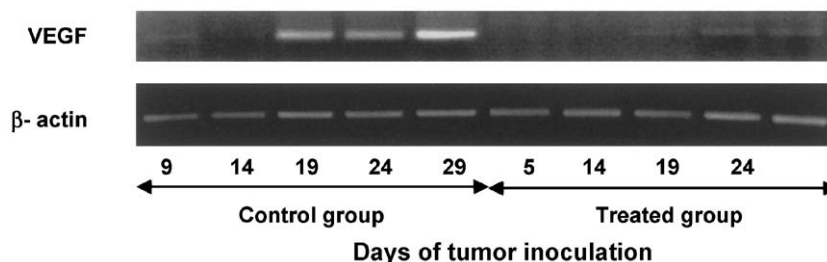


Fig. 7. Gene expression of VEGF ( $\beta$ -actin served as the internal control) in control and treated groups.

the endothelium and opens interendothelial junctions, thus producing leaky hyperpermeable vessels (Dvorak et al., 1995; Roberts and Palade, 1997), features which were amply demonstrated in our SEM pictures (Fig. 4c–d).

Morphometry of the control and PSP-treated corrosion casts showed significant differences in the intervascular distances and mean vessel diameters even though the frequency distribution curves were essentially similar. This and the fact that the interbranch distances showed no significant differences indicate that the tumor-type specific vascular architecture (Konerding et al., 1999) is influenced only to a limited extent by the PSP treatment. In this study we did not assess the branching angles as another parameter for vascular geometry, because our previous studies have not demonstrated any significant differences in a variety of murine tumors and human xenografts (Konerding et al., 1999). Similar small changes in the vascular architectures were also observed in tumors after treatment with FGF-2 (Konerding et al., 1998). However, the correlation between vascular density and tumor weight demonstrated clearly the importance of angiogenesis to tumor growth. The relationship between the area of the blood vessels and the tumor weight was less consistent. This may be due to the shrinkage and collapse of the blood vessels during processing, making the area data less reliable for comparison.

The rationale for assessing microvessel density in selected areas, compared to an overall vascular count, is twofold. From a practical point of view, given the intratumoral heterogeneity, obtaining a reproducible overall assessment of microvessel density requires the evaluation of large tumor areas. From a pathophysiological point of view, it may be expected that the mechanism responsible for the direct and indirect relations between the tumor and the endothelial cell population is particularly evident in highly vascular regions (Vermeulen et al., 1996).

Results from the microcorrosion casting and immunohistochemical studies indicate that the tumor vasculature was closely related to tumor progression, in that the PSP-treated group showed far fewer active angiogenic features. The gene expression study largely confirmed the anti-angiogenic effects of PSP, as the expression of VEGF was much lower in the treated group. Since VEGF is a key factor in angiogenesis, its down-regulation resulted in less stimulation of tumor angiogenesis. Woessner et al. (1997) have shown that tumor regression is related to a continuous decrease of VEGF mRNA expression while tumor growth inhibition is related to a stable mRNA expression. The findings in this study are in agreement with their results, in that tumor growth inhibition, rather than regression, was observed, with a stable and low expression of VEGF mRNA.

The tumors in the untreated and PSP-treated groups differed by 48% in mean weights. Such an inhibitory effect is quite large when compared with the other anti-tumor growth studies with the S180 solid tumor model with respect to dosage used and the percentage of inhibition. A 30–70% inhibition has been reported in previous studies using 1.25 mg/mouse/day polysaccharopeptides isolated from another edible mushroom, *Tricholoma mongolicum* (Wang et al., 1995, 1996). The polysaccharopeptide in this study was given at a dose of 35 µg/mouse/day for 29 days via oral feeding. By the use of recombinant tissue inhibitor of metalloproteinase-3 (rTIMP-3) at a dose of 20 mg/kg body weight for 10 days (i.e. about 500 mg/mouse/day, taking the average weight of the mouse as 25 g), the inhibitory effect was 63.2% (Jianxin et al., 1998). Somatostatin analogue TT-232 produced inhibition up to about 50% when given at 750 mg/kg body weight (i.e. about 18.75 mg/mouse/day) (Keri et al., 1996). Red ginseng extracts A and B produced an inhibitory effect of 47% and 45% respectively, at a dose of 400 mg/kg body weight (i.e. about 10 mg/mouse/day) after 10 days of oral feeding.

The present study demonstrates that the polysaccharopeptide of *Coriolus versicolor* has a significant inhibitory effect on the growth of S180 solid tumors. Evidence at hand indicates that the anti-tumor activity of PSP resides in its anti-angiogenic properties, via a suppression of VEGF gene expression, resulting in a deprivation of angiogenic stimulation to the tumor growth. It remains, however, to determine if other angiogenic factors, such as hepatocytes growth factors (HGF) and its receptor c-Met, urokinase plasminogen activator (uPA) and its receptor uPAR, basic fibroblast growth factor (bFGF), TGF- $\alpha$  and TGF $\beta$ -1, have an active role in tumor growth inhibition. Our data presents yet another piece of evidence to the growing portfolio of natural products and medicinal herbs which are increasingly showing promise of capable of waging effective combat against the ravages of diseases such as cancer and tumors, among others.

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